

Code:

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#include <stdio.h> #include <stdlib.h>

struct n { int data; struct n *next; };

struct n *head = NULL;

void insert(int data) { struct n temp = (struct n)malloc(sizeof(struct n)); temp->data =
data; temp->next = NULL; if (head == NULL) { head = temp; temp->next = head; } else
{ struct n *curr = head; while (curr->next != head) { curr = curr->next; } curr->next = temp;
temp->next = head; }}

void display(struct n *h) { if (h == NULL) return; struct n *temp = h; printf("[h]"); do
{ printf("=>%d", temp->data); temp = temp->next; } while (temp != h); printf("=>[h]\n"); }

int main() { int n; if (scanf("%d", &n) != 1) return 0;

for (int i = 1; i <= n; i++) {
    insert(i);
}

printf("Complete linked_list:\n");
display(head);

struct n *odd_head = NULL, *odd_tail = NULL;
struct n *even_head = NULL, *even_tail = NULL;

for (int i = 1; i <= n; i++) {
    struct n *temp = (struct n*)malloc(sizeof(struct n));
    temp->data = i;
    temp->next = NULL;

    if (i % 2 != 0) {
        if (odd_head == NULL) {
            odd_head = odd_tail = temp;
            temp->next = odd_head;
        } else {
            odd_tail->next = temp;
            temp->next = odd_head;
            odd_tail = temp;
        }
    } else {

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        if (even_head == NULL) {
            even_head = even_tail = temp;
            temp->next = even_head;
        } else {
            even_tail->next = temp;
            temp->next = even_head;
            even_tail = temp;
        }
    }
}

printf("Odd:\n");
display(odd_head);

printf("Even:\n");
display(even_head);

return 0;

}

```

Input:

6

Output:

Complete linked_list:

[h]=>1=>2=>3=>4=>5=>6=>[h]

Odd:

[h]=>1=>3=>5=>[h]

Even:

$$[h] \Rightarrow 2 \Rightarrow 4 \Rightarrow 6 \Rightarrow [h]$$